



HIMVIGYAN

Integrated Polar & Ocean Science Knowledge Platform

National Centre for Polar and Ocean Research (NCPOR) * Ministry of Earth Sciences (MoES)

PROJECT SUMMARY, ARCHITECTURE & TECHNICAL DOSSIER

HimVigyan is an integrated, full-stack digital knowledge ecosystem engineered to centralize, preserve, and disseminate India’s four decades of polar exploration and cryospheric research across Antarctica, the Arctic, the Himalayas, and the Southern Ocean. Designed specifically for SIH Problem Statement 26063, this enterprise-grade platform bridges the gap between deep polar science and public understanding through real-time open datasets, AI-assisted multilingual communication, interactive student gamification, and verified researcher publication workflows.

PROJECT METADATA & SPECIFICATIONS	
Problem Statement ID:	26063 (Ministry of Earth Sciences / NCPOR)
Platform Name:	HimVigyan (Formerly Polar India Portal)
Primary Purpose:	Centralized Polar Science Dissemination, Datasets, Media & Education
Frontend Architecture:	Next.js 16.3.6 (App Router) + React 19.2.8 + TypeScript 5
Styling & Theme Engine:	Tailwind CSS v4 + PostCSS + Dark/Light Theme Persistence
Database & Auth:	Supabase Cloud (PostgreSQL + RLS + Storage Buckets)
Document Version:	Version 1.0.0 (Production Hackathon Release * September 2026)

CORE VALUE PROPOSITIONS

1. Unified Open Archive	2. AI Outreach Studio	3. Student Gamification
Centralizes scientific datasets, official expedition compendiums, research papers, and ice-core registries with verified DOI minting.	Transforms dense polar research papers into PIB press communiques, social threads, and bilingual Hindi outreach ready for publishing.	Interactive 10-question Polar Quiz, confetti animations, Himalayan glacier-melt simulator, and verified digital MoES certificates.

1. Executive Summary & Problem Context

India holds an esteemed four-decade legacy in polar sciences, encompassing three permanent stations in Antarctica (Dakshin Gangotri, Maitri, and Bharati), the high-latitude Arctic station Himadri in Ny-Alesund (Svalbard), the multi-sensor IndARC underwater ocean mooring, the high-altitude Himansh station in the Western Himalayas (Chandra Basin), and recurring Southern Ocean scientific expeditions (ISOE). Despite this formidable scientific output, access to these crucial findings has historically been hindered by fragmented repositories, manual outreach pipelines, and complex formats inaccessible to non-specialists.

Problem Statement 26063 Objectives:

- Centralized Scientific Knowledge Hub: Build a single-window portal where Indian polar expeditions, station telemetry, scientific publications, ice core datasets, and high-resolution media are discoverable and downloadable.
- Bilingual & Multi-Tiered Dissemination: Provide AI-assisted conversion of complex scientific monographs into accessible English and Hindi press releases, social media threads, and educational digests.
- Youth & Citizen Science Engagement: Incorporate interactive cryospheric simulators, real-time quizzes, and authenticated achievement badges to inspire the next generation of polar scholars.
- Authorized Researcher Contribution Pipeline: Enable field scientists to log in, deposit field compendiums, submit metadata, mint verified DOIs, and track citation analytics seamlessly.

2. High-Level Architectural Blueprint

HimVigyan employs a high-performance modern decoupled architecture. The frontend leverages Next.js 16 (App Router) with React 19 for instantaneous hybrid static and dynamic rendering. Server components provide lightning-fast hydration, while client components manage real-time interactivity. The cloud persistence layer utilizes Supabase PostgreSQL with Row Level Security (RLS) policies and S3-compatible cloud storage buckets.

SYSTEM ARCHITECTURE & MULTI-TIER DATA FLOW	
[CLIENT LAYER]	Web Browser / Mobile Viewport (Responsive HTML5, Tailwind v4 Engine) ?
[APPLICATION LAYER]	Next.js 16.3.6 App Router (React 19 Server/Client Components) ??? Navigation & Routing (Home, Expeditions, Research, Media, About, Workflow) ??? State Management (React Hooks + Browser LocalStorage Persistence) ??? Specialized Interactive Micro-Engines (AI Studio, Quiz, Glacier Simulator) ?
[SERVICE LAYER]	Bilingual NLP Generator * Canvas-Confetti FX * Citation Engine (APA/BibTeX) ?
[BACKEND / DATABASE]	Supabase Cloud (PostgreSQL 15 RLS * Supabase Auth * S3 Storage Buckets)

3. Comprehensive Technology Stack

HimVigyan is developed using the latest production-grade open web standards, prioritizing high accessibility, zero layout shifts, type safety, and real-time client performance.

Layer / Subsystem	Technology / Library	Version	Key Architectural Purpose
Framework Runtime	Next.js (App Router)	16.3.6	Server/Client Components, Webpack optimization
User Interface	React / React-DOM	19.2.8	Component-driven reactive UI & smooth transitions
Type Safety	TypeScript	5.x	End-to-end interface typing & compile-time safety
CSS Styling	Tailwind CSS + PostCSS	v4.0.0	Custom CSS variables, dark/light theme engine
Cloud Database	Supabase PostgreSQL	2.109.0	Managed relational tables, RLS policies, metadata
Authentication	Supabase Auth	2.109.0	Secure JWT sessions, researcher credentials
Cloud Storage	Supabase Storage Buckets	v2 API	PDF report storage, high-res photos, datasets
Gamification FX	Canvas Confetti	1.9.4	Particle physics celebratory triggers for quiz badge
Iconography	Lucide React	1.48.0	40+ semantic, accessible SVG vector icons
Code Quality	ESLint + Next Linter	9.x	Strict code style, clean architecture enforcement

Package.json Dependency Audit:

- Core Dependencies: @supabase/supabase-js (^2.109.0), canvas-confetti (^1.9.4), lucide-react (^1.48.0), next (16.3.6), react (19.2.8), react-dom (19.2.8)
- Development Tooling: @tailwindcss/postcss (^4), @types/canvas-confetti (^1.9.0), @types/node (^20), @types/react (^19), @types/react-dom (^19), eslint (^9), eslint-config-next (16.3.6), tailwindcss (^4), typescript (^5)

[+] Zero-Runtime Overhead Design

The application leverages modern CSS variables and Tailwind v4 compiled styles, eliminating runtime CSS-in-JS libraries. Client bundles remain

4. Detailed Module-by-Module Breakdown

Module 1: Unified Header & Navigation System (Navbar.tsx, ThemeToggle.tsx)

- Official Brand Identity: Features the official HimVigyan Crest (Polar Bear mother & cub with snowy mountain backdrop) and tagline EXPLORE * RESEARCH * PRESERVE.
- Smart Navigation Bar: Direct navigation across Home, Expeditions, Research, Publications, Reports, Datasets, Media, and About.
- Theme Toggle Engine: Smooth toggle between Light Glacier Mode and Deep Ocean Dark Mode, persisting preferences in browser localStorage and html attributes.
- Evaluator / Demo Gateway: Provides instant 1-click test sign-in for evaluators without requiring manual database registrations.

Module 2: Hero Section & Dynamic Discovery Engine (Hero.tsx)

- Visual Immersion: Full-bleed polar landscape background with ambient overlays and live video fallback.
- Multi-Category Search Dropdown: Floating search input supporting category scoping across All Categories, Expeditions, Research, Publications, Reports, Datasets, and Media.
- Quick-Action Call-to-Actions: Direct 'Explore' button jumping into research and 'Contribute' button launching the scientist upload gateway.

Module 3: Indian Scientific Expeditions Catalog (ExpeditionsCatalog.tsx)

- Four Polar Domains: Comprehensive coverage of Antarctica (43-IAE, Maitri, Bharati), Arctic (INDARC, Himadri), Southern Ocean (12-ISOE), and Himalayas (Himansh Station).
- Full Operational Profiles: Station coordinates, operating seasons, historical timelines, weather telemetry, lead scientists, and environmental treaties.
- Associated Scientific Datasets: Each expedition directly cross-references research publications and open datasets with real-time download buttons.

Module 4: Integrated Polar Knowledge Repository (KnowledgeRepository.tsx)

- Faceted Filtering: Real-time tabs for All, Expeditions, Research, Publications, Reports, and Datasets.
- DOI Minting & Citations: Every archived item has a unique DOI (e.g. 10.26063/ncpor.icecore.43iae.01) and generates instant APA, BibTeX, and RIS citations.
- Scenic PDF Report Viewer (ReportDetailModal.tsx): Full-screen modal displaying metadata, lead investigators, executive abstracts, methodology, and direct download links.

Module 5: AI Media & Outreach Studio (AiDisseminator.tsx)

Addresses Problem Statement 26063's explicit mandate for automated science dissemination. Takes technical expedition papers and generates ready-to-publish communication collateral across three key formats:

- PIB Press Communique: Formal government-style release with official dateline, headline, quote by Chief Scientist, key findings, and MoES sign-off.
- Social Media (X/Twitter) Thread: Numbered multi-tweet thread with hashtags (#PolarIndia #MoES #NCPOR), character counter, and engagement hooks.
- Instagram Educational Carousel: Slide-by-slide script featuring image visual suggestions, core science concepts, and call-to-actions.
- Bilingual Engine (English / Hindi): 1-click toggle switches generated content between English and Hindi (Rashtriya Dhruviya Gyan Samachar).

Module 6: Smart Education & Student Citizen Hub (StudentPolarHub.tsx)

Gamified outreach center targeting school students and university researchers:

- Interactive 10-Question Quiz: Tests knowledge on Antarctica Treaty 1959, Indian Antarctic Act 2022, permafrost thaw, IndARC moorings, and polar biology.
- Particle Physics Confetti: Triggers canvas-confetti upon completion, rendering a personalized Verified NCPOR Student Polar Badge.
- Glacier Retreat Timeline Slider: Interactive visual slider allowing students to explore ice volume loss in the Chandra Basin (Chhota Shigri & Samudra Tapu glaciers) from 2000 to 2024.

Module 7: Moments from Polar Frontiers (MediaPortal.tsx, MediaHighlights.tsx)

- 4K UHD Multimedia Archive: Categorized library of high-resolution expedition photographs and documentaries.
- Technical Metadata Badges: Displays aspect ratios, resolutions (3840x2160 UHD), author attribution, expedition year, download counters, and community upvotes.

Module 8: HimVigyan AI Assistant (PolarBot.tsx)

- Contextual Scientific RAG: Conversational agent grounded in central NCPOR archives, station telemetry, and scientist profiles.
- Author & Station Query Parsing: Recognizes queries like 'Dr. Meenakshi Reports' or 'Bharati Station Weather' and returns structured scientist profiles and reports.
- Embedded Actions: Allows users to inspect and download PDF reports directly inside the chat interface.

Module 9: Scientist Contribution & Peer-Review Workflow

Authorized portal (ScientistSubmissionModal.tsx) enabling active researchers to contribute new expedition data directly from base stations or home institutions:

- Metadata Tagging: Collects publication title, lead investigator, expedition number, category (Research, Dataset, Report, Media), and abstract.
- Automated DOI Assignment: Generates an official DOI format (e.g. 10.26063/ncpor.submission...) upon submission.
- Dual-Mode Persistence: When Supabase is configured, uploads metadata to PostgreSQL tables and files to cloud storage; falls back seamlessly to browser localStorage with full CRUD capability.
- Immediate Live Sync: Submitted items immediately reflect across the Knowledge Repository, AI Disseminator, and author profiles without requiring page reloads.

Module 10: About HimVigyan Portal (AboutPortal.tsx)

Executive overview page structured according to the latest government design guidelines:

- Panoramic Antarctic Hero: Featuring 4K imagery of Emperor penguins on sea ice with golden horizon.
- Our Mission & Our Vision: Card-based statement of public outreach and high-latitude scientific stewardship.
- Seven Service Modules: Interactive cards leading directly to Repository, Expeditions, AI Studio, Student Hub, Media, Scientist Submission, and AI Bot.
- Four Focus Regions: Circular high-definition badges covering Antarctica, the Arctic, the Himalayas, and the Southern Ocean.
- Ethical Commitment Banner: Reaffirming scientific integrity, source attribution, data provenance, and fair open access.

Module 11: End-to-End Workflow Slide (app/workflow/page.tsx)

- 16:9 Presentation Format: Built specifically for SIH presentation slides and evaluator walkthroughs.
- Four Interactive Zones: Zone 1 (Public User Discovery), Zone 2 (Authorized Researcher), Zone 3 (Institutional Review & Moderation), and Zone 4 (Technical Architecture & Cloud Storage).

5. Data Architecture & Schema Models

HimVigyan maintains a strictly typed data model matching the International Polar Year (IPY) and Antarctic Treaty System metadata requirements:

Field Name	TypeScript Type	Description / Constraints	Example Value
id	string	Unique identifier for repository item	'rep-01'
title	string	Official paper or dataset title	'43-IAE Larsemann Hills Ice Core'
type	string	'Research' 'Dataset' 'Report' 'Media'	'Dataset'
category	string	Expedition topic classification	'Glaciology & Cryosphere'
date	string	Publication or collection date	'March 2024'
author	string	Principal Investigator name & title	'Dr. Thamban Meloth'
doi	string	Standard Digital Object Identifier	'10.26063/ncpor.icecore.43iae.01'
station	string	Associated research base or vessel	'Bharati Station, Antarctica'
downloadCount	number	Aggregate public access metric	1420

6. Security, Privacy & Performance Standards

- Row-Level Security (RLS): PostgreSQL tables in Supabase restrict update/delete permissions strictly to authenticated researchers and administrators, while granting public read access.
- WCAG 2.1 Accessibility: High-contrast text colors, semantic HTML5 tags (<header>, <nav>, <main>, <section>, <footer>), and ARIA attributes ensure full compliance.
- Hydration Protection: Inline theme script in app/layout.tsx reads user theme preference prior to DOM rendering, eliminating theme flickering and hydration errors.
- Offline-First Resilience: If internet connectivity or Supabase is temporarily unreachable, all client functions (quiz, glacier simulator, AI preview, and search) continue operating smoothly using local caches.

7. Competitive Advantage Matrix (SIH Evaluation)

Comparison between legacy polar websites and HimVigyan's modern architecture:

Evaluation Metric	Legacy MoES / Polar Sites	HimVigyan Platform (Problem 26063)
Data Discoverability	Scattered PDF links across static pages	Faceted multi-category search with DOI minting
AI Science Outreach	Manual press releases (delays of weeks)	Instant PIB, X threads & Hindi carousels
Student Engagement	Static text and brochures	10-Q Quiz, Confetti, Glacier Timeline & Certificates
Field Submissions	Paper forms or unindexed email attachme..	Direct web portal with metadata tagging & instant sync
Modern UX / Theme	Non-responsive, fixed-width desktop lay..	Fully responsive Tailwind v4 with dark/light themes
Offline Resilience	Fails completely without active network	LocalStorage fallbacks and zero-downtime demo mode

8. Future Production Scaling Milestones

- Phase 1 (Immediate Production): Deploy on Vercel / National Informatics Centre (NIC) Cloud with custom government domain (himvigyan.moes.gov.in).
- Phase 2 (IoT & Satellite Integration): Direct automated ingestion of Bharati & Maitri AWS (Automatic Weather Station) telemetry via GSAT/Inmarsat satellite relays.
- Phase 3 (3D Virtual Station Tours): WebGL-based 3D digital twins of Bharati Station interior and Ny-Alesund atmospheric laboratory.
- Phase 4 (National School Challenge): Annual nationwide Polar Science Quiz competition linked with NCPOR student expedition fellowship grants.

HIMVIGYAN * EXPLORE * RESEARCH * PRESERVE

Submitted for Smart India Hackathon 2026 Evaluation * Ministry of Earth Sciences, Govt. of India